# Annotated Bangla/Bengali Speech Corpora — Reference Overview

A structured summary of eight major Bangla/Bengali speech corpora, with citation details and links for verification.

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## 1. Annotated Bangla Speech Corpora (CRBLP)

**Citation:** @article{alam2010development, title={Development of annotated bangla speech corpora}, author={Alam, Firoj and Habib, SM and Sultana, Dil Afroza and Khan, Mumit}, year={2010}, publisher={BRAC University} }

**DOI:** Not assigned (ISCA workshop archive paper, no DOI issued)

**Link:** https://www.isca-archive.org/sltu_2010/alam10_sltu.pdf

**Corpus repository:** https://github.com/Bangla-Language-Processing/Bangla-Speech-Corpora

| Attribute | Details |
|---|---|
| Dataset name | Bangla annotated read speech corpora (Corpus for acoustic analysis, Diphone corpus, Continuous speech corpus) |
| Size / recording length | 13 hrs 32 min cleaned (from 24 hrs raw) |
| Total sentences | ~10,000 (10,895 continuous corpus; 4,335 diphone corpus; 262 acoustic corpus) |
| Sampling rate | 44.1 kHz, 24-bit |
| Recording setup | Studio recording, Tascam TM-D4000 Digital Mixer + Audio-Technica microphone |
| Speakers | Native speakers — professional voice talents and non-professionals (students) |
| Speaker distribution | Acoustic corpus: 4 male + 4 female (professional 52–54 yrs, non-professional 25–29 yrs); Diphone/Continuous corpora: single 29-year-old professional male |
| Bangla variant | Bangladeshi Bangla (West Bengal speakers excluded) |
| Open sourced | Yes |

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|---|---|
| Parallel corpus | Not Parallel |
| Purpose | Phonetic research, acoustic analysis, TTS, STT |

## 2. SHRUTI Bengali Continuous ASR Speech Corpus

**Citation:** @inproceedings{das2011bengali, title={Bengali speech corpus for continuous auutomatic speech recognition system}, author={Das, Biswajit and Mandal, Sandipan and Mitra, Pabitra}, booktitle={2011 International conference on speech database and assessments (Oriental COCOSDA)}, pages={51--55}, year={2011}, organization={IEEE} }

**DOI:** 10.1109/ICSDA.2011.6085979

**Link:** https://ieeexplore.ieee.org/document/6085979

| Attribute | Details |
|---|---|
| Dataset name | SHRUTI |
| Size / recording length | 26 hours continuous read speech (alt. figure cited: 21.64 hours) |
| Total sentences | 7,500 unique sentences; 19,640 unique words |
| Sampling rate | 16 kHz, mono, 16-bit |
| Recording setup | Room environment, Sony FV-220 microphone, Emu speech tool |
| Speakers | Native speakers: younger (20–40 yrs) and older (60–80 yrs) groups |
| Speaker distribution | 110 speakers (70 male, 40 female): 30 young M, 40 older M, 20 young F, 20 older F |
| Bangla variant | Standard colloquial Bengali, mostly West Bengal (Rarh dialect, south-western region) |
| Open sourced | Not Found |
| Parallel corpus | Not Parallel |
| Purpose | Speaker-independent continuous ASR (STT) |

## 3. Large Bengali ASR Training Data (OpenSLR SLR53)

**Citation:** @inproceedings{kjartansson2018crowd, title={Crowd-Sourced Speech Corpora for Javanese, Sundanese, Sinhala, Nepali, and Bangladeshi Bengali.}, author={Kjartansson, Oddur

and Sarin, Supheakmungkol and Pipatsrisawat, Knot and Jansche, Martin and Ha, Linne}, booktitle={SLTU}, pages={52--55}, year={2018} }

**DOI:** Not assigned (workshop dataset paper hosted on OpenSLR; no DOI issued)

**Link:** https://openslr.org/53/

**Paper (ISCA archive):** https://www.isca-archive.org/sltu_2018/index.html

| Attribute | Details |
|---|---|
| Dataset name | Large Bengali ASR training data (OpenSLR SLR53) |
| Size / recording length | 229 hours, 232,537 audio files |
| Total sentences | ~196k unique utterances |
| Sampling rate | 16 kHz, 16-bit linear PCM |
| Recording setup | Crowd-sourced via DataHound tool on smartphones, quiet environments |
| Speakers | Native-speaker volunteers |
| Speaker distribution | 508 speakers (gender mix unspecified) |
| Bangla variant | Bangladeshi Bengali |
| Open sourced | Yes — CC BY-SA 3.0 US |
| Parallel corpus | Not Parallel |
| Purpose | Acoustic modeling for ASR (STT) |

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## 4. SUBAK.KO (সুবাক্য)

**Citation:** @article{kibria2022bangladeshi, title={Bangladeshi Bangla speech corpus for automatic speech recognition research}, author={Kibria, Shafkat and Samin, Ahnaf Mozib and Kobir, M Humayon and Rahman, M Shahidur and Selim, M Reza and Iqbal, M Zafar}, journal={Speech Communication}, volume={136}, pages={84--97}, year={2022}, publisher={Elsevier} }

**DOI:** 10.1016/j.specom.2021.12.004

**Link:** https://www.sciencedirect.com/science/article/abs/pii/S0167639321001370

**Corpus Link:** https://huggingface.co/datasets/SUST-CSE-Speech/SUBAK.KO

| Attribute | Details |
|---|---|
| Dataset name | SUBAK.KO (সুবাক্য) |
| Size / recording length | 241 hours total (229 hrs recorded + 12 hrs 19 min broadcasted) |
| Total sentences | 77,643 audio files with transcriptions |
| Sampling rate | 16 kHz (downsampled from 48 kHz) |
| Recording setup | Mixed: studio recordings (Audacity + M-Track interface) + broadcasted speech (YouTube, Facebook) |
| Speakers | 61 native-speaker volunteers + 2 L2 speakers (Germany, Nepal) |
| Speaker distribution | 33 male, 28 female, aged 18–35, covering all 8 divisions / 34 districts of Bangladesh |
| Bangla variant | Bangladeshi Bangla — standard colloquial (SCBB) plus regional dialects (e.g., Sylheti) |
| Open sourced | Yes |
| Parallel corpus | Not Parallel |
| Purpose | Large Vocabulary Continuous Speech Recognition (LVCSR), regional-accent ASR research |

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## 5. Bengali Common Voice (v9.0) Speech Dataset

**Citation:** @article{alam2022bengali, title={Bengali common voice speech dataset for automatic speech recognition}, author={Alam, Samiul and Sushmit, Asif and Abdullah, Zaowad and Nakkhatra, Shahrin and Ansary, MD and Hossen, Syed Mobassir and Mehnaz, Sazia Morshed and Reasat, Tahsin and Humayun, Ahmed Imtiaz}, journal={arXiv preprint arXiv:2206.14053}, year={2022} }

**DOI/arXiv ID:** arXiv:2206.14053

**Link:** https://arxiv.org/abs/2206.14053

| Attribute | Details |
|---|---|
| Dataset name | Bengali Common Voice Speech Dataset |
| Size / recording length | 399 hours |
| Total sentences | 231,120 clips |

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|---|---|
| Sampling rate | 48 kHz |
| Recording setup | Crowdsourced via Mozilla Common Voice platform, uncontrolled environments |
| Speakers | 19,817 contributors |
| Speaker distribution | 84.3% male, 15.7% female; 28 self-reported accents (Bangladesh + India) |
| Bangla variant | General Bengali, diverse regional accents |
| Open sourced | Yes — CC-0 Public Domain |
| Parallel corpus | Not Parallel |
| Purpose | Sentence-level ASR (STT) |

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## 6. OOD-Speech

**Citation:** @article{rakib2023ood, title={Ood-speech: A large bengali speech recognition dataset for out-of-distribution benchmarking}, author={Rakib, Fazle Rabbi and Dip, Souhardya Saha and Alam, Samiul and Tasnim, Nazia and Shihab, Md Istiak Hossain and Ansary, Md Nazmuddoha and Hossen, Syed Mobassir and Meghla, Marsia Haque and Mamun, Mamunur and Sadeque, Farig and others}, journal={arXiv preprint arXiv:2305.09688}, year={2023} }

**DOI/arXiv ID:** arXiv:2305.09688

**Link:** https://arxiv.org/abs/2305.09688

**Project page:** https://bengaliai.github.io/asr

| Attribute | Details |
|---|---|
| Dataset name | OOD-Speech |
| Size / recording length | 1,201.33 hours total (Training: 1,129.46 hrs; Test: 23.03 hrs) |
| Total sentences | ~934,000 samples (training set) |
| Sampling rate | Scripted data mostly 48 kHz (Mozilla platform); spontaneous data i rwav |
| Recording setup | Massively crowdsourced (Mozilla Common Voice, Shobdo platforms) + YouTube-sourced spontaneous speech across 17 domains |
| Speakers | 22,645+ native Bengali speakers |

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|---|---|
| Speaker distribution | South Asian native speakers; 28 unique self-reported dialects/accents |
| Bangla variant | Bengali, high regional/prosodic diversity |
| Open sourced | Yes — CC BY-SA 3.0 US |
| Parallel corpus | Not Parallel |
| Purpose | First out-of-distribution (OOD) benchmarking dataset for Bengali ASR |

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## 7. SUST Bangla Emotional Speech Corpus (SUBESCO)

**Citation:** @article{sultana2021sust, title={SUST Bangla Emotional Speech Corpus (SUBESCO): An audio-only emotional speech corpus for Bangla}, author={Sultana, Sadia and Rahman, M Shahidur and Selim, M Reza and Iqbal, M Zafar}, journal={Plos one}, volume={16}, number={4}, pages={e0250173}, year={2021}, publisher={Public Library of Science San Francisco, CA USA} }

**DOI:** 10.1371/journal.pone.0250173

**Link:** https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0250173

**Data repository:** https://doi.org/10.5281/zenodo.4526477

| Attribute | Details |
|---|---|
| Dataset name | SUBESCO |
| Size / recording length | 7 hrs 40 min 40 sec |
| Total sentences | 7,000 recordings (10 sentences × 7 emotions × 20 speakers × 5 repetitions) |
| Sampling rate | 48 kHz, 32-bit mono .wav |
| Recording setup | Acted/simulated; anechoic studio, Wharfedale Pro condenser microphone |
| Speakers | 20 professional speakers/actors |
| Speaker distribution | Gender-balanced (10 male, 10 female), aged 21–35 |
| Bangla variant | Standard Bangla |
| Open sourced | Yes — CC BY 4.0 |
| Parallel corpus | Yes |

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|---|---|
| Purpose | Speech Emotion Recognition (SER), emotional TTS, phonetic/prosodic analysis |

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## 8. BanglaSER

**Citation:** @article{das2022banglaser, title={BanglaSER: A speech emotion recognition dataset for the Bangla language}, author={Das, Rakesh Kumar and Islam, Nahidul and Ahmed, Md Rayhan and Islam, Salekul and Shatabda, Swakkhar and Islam, AKM Muzahidul}, journal={Data in Brief}, volume={42}, pages={108091}, year={2022}, publisher={Elsevier} }

**DOI:** 10.1016/j.dib.2022.108091

**Link:** https://www.sciencedirect.com/science/article/pii/S2352340922004616

**Data repository:** https://data.mendeley.com/datasets/t9h6p943xy/5

| Attribute | Details |
|---|---|
| Dataset name | BanglaSER |
| Size / recording length | ~1 hr 29 min (1,467 audio clips) |
| Total sentences | 1,467 recordings using 3 target sentences |
| Sampling rate | 44.1 kHz, 32-bit float .wav |
| Recording setup | Acted/enacted; smartphones, laptops, external microphones (personal devices) |
| Speakers | 34 non-professional actors |
| Speaker distribution | Balanced (17 male, 17 female), aged 19–47, based in Dhaka, Bangladesh |
| Bangla variant | Bangla with Bengali accent |
| Open sourced | Yes — Mendeley Data, CC BY license |
| Parallel corpus | Yes |
| Purpose | Speech Emotion Recognition (SER), ML model training |

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### Notes on DOI/link verification

- Papers **1** (CRBLP) and **3** (OpenSLR SLR53) are workshop papers (SLTU 2010/2018) that were

never assigned a formal DOI — they're archived by ISCA and OpenSLR respectively, so the direct archive/dataset link is provided instead.

- Paper **2** (SHRUTI) has a genuine IEEE DOI since it was an IEEE Oriental COCOSDA conference paper.
- Papers **5** (Bengali Common Voice) and **6** (OOD-Speech) are arXiv preprints; arXiv IDs are given in place of a DOI since no journal DOI has been assigned.
- Papers **4** (SUBAK.KO), **7** (SUBESCO), and **8** (BanglaSER) have full journal DOIs (Speech Communication, PLOS ONE, and Data in Brief respectively).